



PORTFOLIO OPTIMIZATION & RISK ASSESSMENT

Group 1 - Project 3

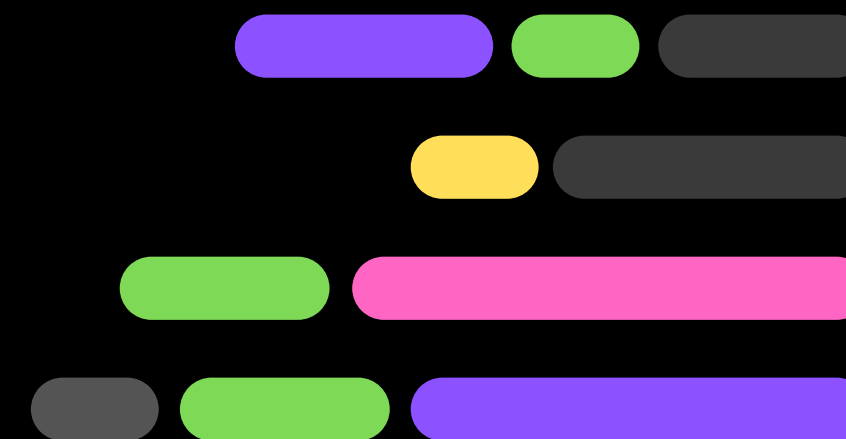
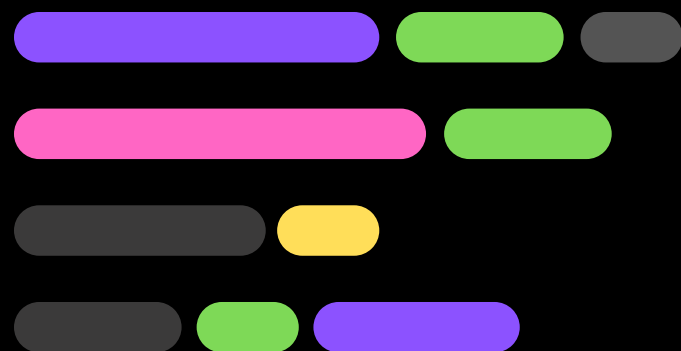
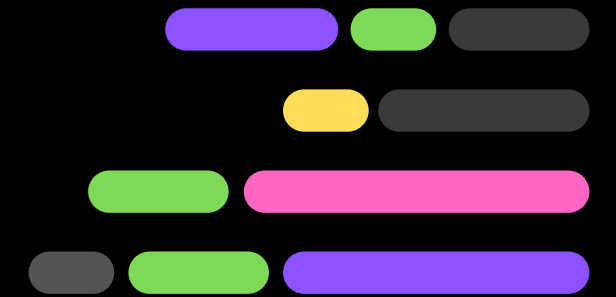


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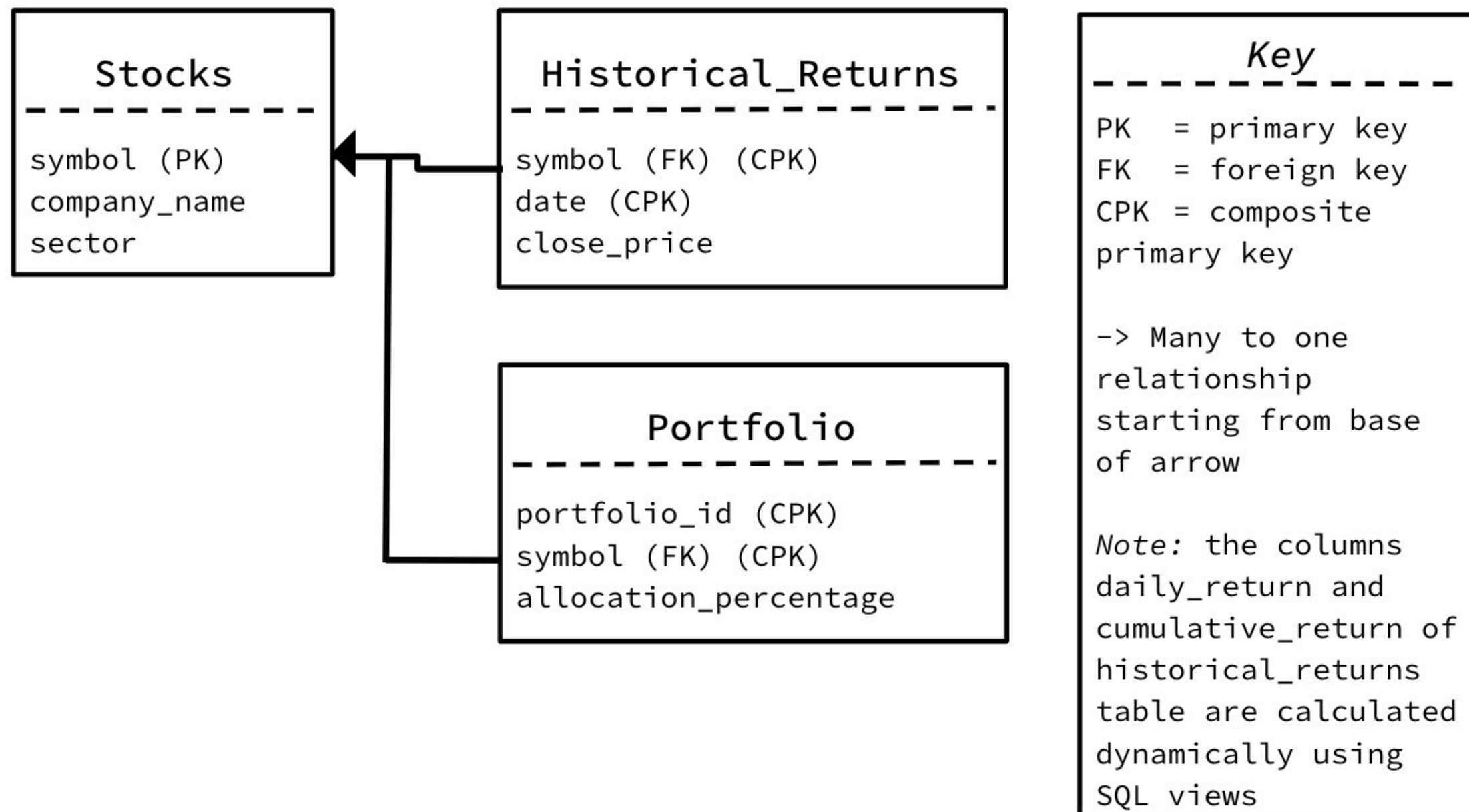
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02

Schema Creation Script

```
CREATE DATABASE `portfolio_optimization` /*!40100 DEFAULT  
CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci */ /*!80016  
DEFAULT ENCRYPTION='N' */;
```



02

Stock Table Setup - 1/9

Table Creation & Insertion queries for first table :

```
CREATE TABLE stocks (  
    symbol VARCHAR(10) NOT NULL PRIMARY KEY,  
    company_name VARCHAR(100),  
    sector VARCHAR(50)  
);
```

```
INSERT INTO stocks (symbol, company_name, sector) VALUES  
( 'AAPL', 'Apple Inc.', 'Technology'),  
( 'MSFT', 'Microsoft Corporation', 'Technology'),  
( 'TSLA', 'Tesla Inc.', 'Automotive/Technology'),  
( 'GOOGL', 'Alphabet Inc. (Class A)', 'Communication  
Services'),  
( 'GOOG', 'Alphabet Inc. (Class C)', 'Communication  
Services'),  
( 'AMZN', 'Amazon.com Inc.', 'Consumer Discretionary'),  
( 'META', 'Meta Platforms Inc.', 'Communication Services'),  
( 'NVDA', 'NVIDIA Corporation', 'Technology'),  
( 'JPM', 'JPMorgan Chase & Co.', 'Financial'),  
( 'BAC', 'Bank of America Corp.', 'Financial'),  
( 'CVX', 'Chevron Corporation', 'Energy'),  
( 'XOM', 'Exxon Mobil Corporation', 'Energy'),  
( 'UNH', 'UnitedHealth Group', 'Healthcare'),  
( 'GM', 'General Motors', 'Automotive'),  
( 'AVGO', 'Broadcom Inc.', 'Technology'),  
( 'BRK-B', 'Berkshire Hathaway (B)', 'Financial'),  
( 'NFLX', 'Netflix Inc.', 'Communication Services'),  
( 'DIS', 'The Walt Disney Company', 'Communication  
Services'),  
( 'PEP', 'PepsiCo Inc.', 'Consumer Staples'),  
( 'WMT', 'Walmart Inc.', 'Consumer Staples');
```


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Stock Table Screenshots

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	symbol	company_name	sector
▶	AAPL	Apple Inc.	Technology
	AMZN	Amazon.com, Inc.	Consumer Discretionary
	AVGO	Broadcom Inc.	Technology
	BAC	Bank of America	Financial
	BRK-B	Berkshire Hathaway Inc. (Class B)	Financial
	CVX	Chevron Corp.	Energy
	DIS	The Walt Disney Company	Communication Services
	GM	General Motors	Automotive
	GOOG	Alphabet Inc. (Class C)	Technology
	GOOGL	Alphabet Inc. (Class A)	Technology
	JPM	JPMorgan Chase	Financial
	META	Meta Platforms, Inc.	Technology
	MSFT	Microsoft Corporation	Technology

Result Grid				Filter Rows:	Edit:
	symbol	company_name	sector		
	GOOG	Alphabet Inc. (Class C)	Technology		
	GOOGL	Alphabet Inc. (Class A)	Technology		
	JPM	JPMorgan Chase	Financial		
	META	Meta Platforms, Inc.	Technology		
	MSFT	Microsoft Corporation	Technology		
	NFLX	Netflix, Inc.	Communication Services		
	NVDA	NVIDIA Corporation	Technology		
	PEP	PepsiCo, Inc.	Consumer Staples		
	TSLA	Tesla, Inc.	Automotive		
	UNH	UnitedHealth Group	Healthcare		
	WMT	Walmart Inc.	Consumer Staples		
	XOM	Exxon Mobil	Energy		



02

Create Historical Return Table - 3/9

Script for creating historical_returns table :

```
CREATE TABLE historical_returns (  
    symbol VARCHAR(10),  
    date DATE,  
    close_price DECIMAL(10, 2),  
    PRIMARY KEY (symbol, date),  
    FOREIGN KEY (symbol) REFERENCES stocks(symbol)  
);
```




```
import yfinance as yf
import csv
import pandas as pd

stocks = [
    "AAPL", "MSFT", "TSLA", "GOOGL", "GOOG", "AMZN", "META", "NVDA",
    "JPM", "BAC", "CVX", "XOM", "UNH", "GM", "AVGO", "BRK-B",
    "NFLX", "DIS", "PEP", "WMT"
]

start_date = "2024-01-02"
end_date = "2024-06-30"
```

Filling Data in Historical Returns Table Using Python Script - 4/9

```
with open("raw_prices_20stocks.csv", "w", newline="") as csvfile:
    csvwriter = csv.writer(csvfile)
    csvwriter.writerow(["symbol", "date", "close_price"])

total_rows = 0
for stock in stocks:
    data = yf.download(stock, start=start_date, end=end_date, progress=False)
    if not data.empty:
        close_data = data["Close"]
        for date_index in close_data.index:
            price = close_data.loc[date_index]
            if hasattr(date_index, 'strftime'):
                date_str = date_index.strftime("%Y-%m-%d")
            else:
                try:
                    date_str = pd.to_datetime(date_index).strftime("%Y-%m-%d")
                except:
                    date_str = str(date_index)

            close_price = round(float(price), 2)

            csvwriter.writerow([stock, date_str, close_price])
            total_rows += 1
```





02

Historical Table Data -

5/9

Loading data into historic_return table :

```
LOAD DATA INFILE 'path/to/file.csv'  
INTO TABLE historical_returns  
FIELDS TERMINATED BY ','  
ENCLOSED BY '"'  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS  
(symbol, date, close_price);
```



02

Historical Return Table Calculations - 6/9

Calculating cumulative and daily return data calculations :

```
CREATE OR REPLACE VIEW view_daily_returns AS
SELECT
    symbol,
    date,
    close_price,
    ROUND((close_price / LAG(close_price) OVER (PARTITION BY symbol ORDER BY date)) - 1, 4) AS daily_return
FROM historical_returns;
```

```
CREATE OR REPLACE VIEW view_cumulative_returns AS
SELECT
    symbol,
    date,
    close_price,
    ROUND(EXP(SUM(LOG(1 + daily_return)) OVER (PARTITION BY symbol ORDER BY date)), 6) AS cumulative_return
FROM view_daily_returns
WHERE daily_return IS NOT NULL;
```

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Historical Return Table

Calculations Table - 7/9

	symbol	date	close_price	daily_return
▶	AAPL	2024-01-02	184.53	NULL
	AAPL	2024-01-03	183.15	-0.0075
	AAPL	2024-01-04	180.82	-0.0127
	AAPL	2024-01-05	180.10	-0.0040
	AAPL	2024-01-08	184.45	0.0242
	AAPL	2024-01-09	184.04	-0.0022
	AAPL	2024-01-10	185.08	0.0057
	AAPL	2024-01-11	184.48	-0.0032
	AAPL	2024-01-12	184.81	0.0018
	AAPL	2024-01-16	182.53	-0.0123
	AAPL	2024-01-17	181.59	-0.0051
	AAPL	2024-01-18	187.50	0.0325
	AAPL	2024-01-19	190.42	0.0156
	AAPL	2024-01-22	192.73	0.0121
	AAPL	2024-01-23	194.02	0.0067
	AAPL	2024-01-24	193.34	-0.0035
	AAPL	2024-01-25	193.01	-0.0017

	symbol	date	close_price	daily_return
	AMZN	2024-01-02	149.93	NULL
	AMZN	2024-01-03	148.47	-0.0097
	AMZN	2024-01-04	144.57	-0.0263
	AMZN	2024-01-05	145.24	0.0046
	AMZN	2024-01-08	149.10	0.0266
	AMZN	2024-01-09	151.37	0.0152
	AMZN	2024-01-10	153.73	0.0156
	AMZN	2024-01-11	155.18	0.0094
	AMZN	2024-01-12	154.62	-0.0036
	AMZN	2024-01-16	153.16	-0.0094
	AMZN	2024-01-17	151.71	-0.0095
	AMZN	2024-01-18	153.50	0.0118
	AMZN	2024-01-19	155.34	0.0120
	AMZN	2024-01-22	154.78	-0.0036
	AMZN	2024-01-23	156.02	0.0080
	AMZN	2024-01-24	156.87	0.0054
	AMZN	2024-01-25	157.75	0.0056

	symbol	date	close_price	daily_return
	AVGO	2024-01-02	106.77	NULL
	AVGO	2024-01-03	104.14	-0.0246
	AVGO	2024-01-04	103.19	-0.0091
	AVGO	2024-01-05	103.22	0.0003
	AVGO	2024-01-08	105.74	0.0244
	AVGO	2024-01-09	106.49	0.0071
	AVGO	2024-01-10	106.30	-0.0018
	AVGO	2024-01-11	108.21	0.0180
	AVGO	2024-01-12	108.97	0.0070
	AVGO	2024-01-16	109.68	0.0065
	AVGO	2024-01-17	108.57	-0.0101
	AVGO	2024-01-18	112.53	0.0365
	AVGO	2024-01-19	119.15	0.0588
	AVGO	2024-01-22	120.06	0.0076
	AVGO	2024-01-23	120.64	0.0048
	AVGO	2024-01-24	123.35	0.0225
	AVGO	2024-01-25	121.00	-0.0191

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Historical Return Table

Calculations Table - 8/9

	symbol	date	close_price	cumulative_return
▶	AAPL	2024-01-03	183.15	0.9925
	AAPL	2024-01-04	180.82	0.979895
	AAPL	2024-01-05	180.10	0.975976
	AAPL	2024-01-08	184.45	0.999594
	AAPL	2024-01-09	184.04	0.997395
	AAPL	2024-01-10	185.08	1.00308
	AAPL	2024-01-11	184.48	0.99987
	AAPL	2024-01-12	184.81	1.00167
	AAPL	2024-01-16	182.53	0.98935
	AAPL	2024-01-17	181.59	0.984304
	AAPL	2024-01-18	187.50	1.016294
	AAPL	2024-01-19	190.42	1.032148
	AAPL	2024-01-22	192.73	1.044637
	AAPL	2024-01-23	194.02	1.051636
	AAPL	2024-01-24	193.34	1.047955
	AAPL	2024-01-25	193.01	1.046174
	AAPL	2024-01-26	191.27	1.036758

	symbol	date	close_price	cumulative_return
	AMZN	2024-01-03	148.47	0.9903
	AMZN	2024-01-04	144.57	0.964255
	AMZN	2024-01-05	145.24	0.968691
	AMZN	2024-01-08	149.10	0.994458
	AMZN	2024-01-09	151.37	1.009574
	AMZN	2024-01-10	153.73	1.025323
	AMZN	2024-01-11	155.18	1.034961
	AMZN	2024-01-12	154.62	1.031235
	AMZN	2024-01-16	153.16	1.021542
	AMZN	2024-01-17	151.71	1.011837
	AMZN	2024-01-18	153.50	1.023777
	AMZN	2024-01-19	155.34	1.036062
	AMZN	2024-01-22	154.78	1.032332
	AMZN	2024-01-23	156.02	1.040591
	AMZN	2024-01-24	156.87	1.04621
	AMZN	2024-01-25	157.75	1.052069
	AMZN	2024-01-26	159.12	1.061222

	symbol	date	close_price	cumulative_return
	AVGO	2024-01-03	104.14	0.9754
	AVGO	2024-01-04	103.19	0.966524
	AVGO	2024-01-05	103.22	0.966814
	AVGO	2024-01-08	105.74	0.990404
	AVGO	2024-01-09	106.49	0.997436
	AVGO	2024-01-10	106.30	0.995641
	AVGO	2024-01-11	108.21	1.013562
	AVGO	2024-01-12	108.97	1.020657
	AVGO	2024-01-16	109.68	1.027291
	AVGO	2024-01-17	108.57	1.016916
	AVGO	2024-01-18	112.53	1.054033
	AVGO	2024-01-19	119.15	1.11601
	AVGO	2024-01-22	120.06	1.124492
	AVGO	2024-01-23	120.64	1.129889
	AVGO	2024-01-24	123.35	1.155312
	AVGO	2024-01-25	121.00	1.133246
	AVGO	2024-01-26	118.53	1.110127

02

Portfolio Table Creation & Insertion - 9/9

```
CREATE TABLE portfolio (  
  portfolio_id INT,  
  symbol VARCHAR(10),  
  allocation_percentage DECIMAL(5,2),  
  COMPOSITE PRIMARY KEY (portfolio_id, symbol),  
  FOREIGN KEY (symbol) REFERENCES stocks(symbol)  
);
```

```
INSERT INTO portfolio (portfolio_id, symbol, allocation_percentage) VALUES
```

```
-- Portfolio 1
```

```
(1, 'AAPL', 15.00),  
(1, 'MSFT', 12.00),  
(1, 'TSLA', 10.00),  
(1, 'AMZN', 10.00),  
(1, 'NVDA', 10.00),  
(1, 'GOOGL', 9.00),  
(1, 'META', 8.00),  
(1, 'JPM', 8.00),  
(1, 'CVX', 9.00),  
(1, 'UNH', 9.00),
```

```
-- Portfolio 2
```

```
(2, 'BAC', 10.00),  
(2, 'XOM', 10.00),  
(2, 'GM', 10.00),  
(2, 'AVGO', 10.00),  
(2, 'BRK-B', 10.00),  
(2, 'NFLX', 10.00),  
(2, 'DIS', 10.00),  
(2, 'PEP', 10.00),  
(2, 'WMT', 10.00),  
(2, 'GOOG', 10.00);
```

	portfolio_id	symbol	allocation_percentage
▶	1	AAPL	15.00
	1	AMZN	10.00
	1	CVX	9.00
	1	GOOGL	9.00
	1	JPM	8.00
	1	META	8.00
	1	MSFT	12.00
	1	NVDA	10.00
	1	TSLA	10.00
	1	UNH	9.00
	2	AVGO	10.00
	2	BAC	10.00
	2	BRK-B	10.00
	2	DIS	10.00
	2	GM	10.00
	2	GOOG	10.00
	2	NFLX	10.00
	2	PEP	10.00
	2	WMT	10.00
	2	XOM	10.00



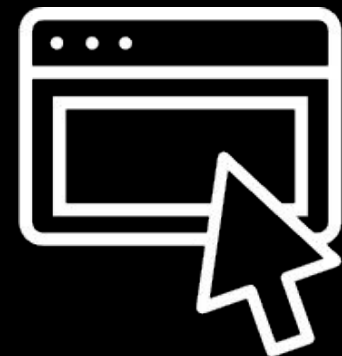
03

SQL Queries



Query 1:

Portfolio returns
based on different
stock weightings.



Query 2:

The best-performing
sector over time.



Query 3:

Stocks with the
lowest risk
(measured by return
volatility).

Query #1 (Daily Portfolio Return Based on Weighted Stock Allocations)

```
CREATE OR REPLACE VIEW view_portfolio_returns AS
WITH stock_returns AS (
    SELECT
        symbol,
        date,
        (close_price - LAG(close_price) OVER (PARTITION BY symbol ORDER BY date)) /
        LAG(close_price) OVER (PARTITION BY symbol ORDER BY date) AS daily_return
    FROM historical_returns
),
weighted_returns AS (
    SELECT
        sr.date,
        sr.symbol,
        sr.daily_return,
        p.portfolio_id,
        sr.daily_return * (p.allocation_percentage / 100) AS weighted_return
    FROM stock_returns sr
    JOIN portfolio p ON sr.symbol = p.symbol
)
SELECT
    portfolio_id,
    date,
    ROUND(SUM(weighted_return), 4) AS portfolio_daily_return
FROM weighted_returns
GROUP BY portfolio_id, date
ORDER BY portfolio_id, date;
```

```
SELECT * FROM view_portfolio_returns
WHERE portfolio_id = 1;
```

```
SELECT * FROM view_portfolio_returns
WHERE portfolio_id = 2;
```


Query #1 Screenshots / Photos

	portfolio_id	date	portfolio_daily_return
▶	1	2024-01-02	NULL
	1	2024-01-03	-0.0056
	1	2024-01-04	-0.0056
	1	2024-01-05	0.0015
	1	2024-01-08	0.0190
	1	2024-01-09	-0.0006
	1	2024-01-10	0.0096
	1	2024-01-11	-0.0008
	1	2024-01-12	-0.0037
	1	2024-01-16	-0.0034
	1	2024-01-17	-0.0048
	1	2024-01-18	0.0084
	1	2024-01-19	0.0121
	1	2024-01-22	0.0007
	1	2024-01-23	0.0041
	1	2024-01-24	0.0072
	1	2024-01-25	-0.0084

	portfolio_id	date	portfolio_daily_return
▶	2	2024-01-02	NULL
	2	2024-01-03	-0.0017
	2	2024-01-04	-0.0049
	2	2024-01-05	0.0018
	2	2024-01-08	0.0091
	2	2024-01-09	-0.0046
	2	2024-01-10	-0.0008
	2	2024-01-11	0.0003
	2	2024-01-12	0.0016
	2	2024-01-16	-0.0037
	2	2024-01-17	-0.0085
	2	2024-01-18	0.0092
	2	2024-01-19	0.0127
	2	2024-01-22	0.0036
	2	2024-01-23	0.0056
	2	2024-01-24	0.0132
	2	2024-01-25	0.0132

	portfolio_id	date	portfolio_daily_return
	1	2024-01-26	0.0008
	1	2024-01-29	0.0117
	1	2024-01-30	-0.0031
	1	2024-01-31	-0.0225
	1	2024-02-01	0.0106
	1	2024-02-02	0.0344
	1	2024-02-05	-0.0030
	1	2024-02-06	0.0023
	1	2024-02-07	0.0126
	1	2024-02-08	0.0004
	1	2024-02-09	0.0105
	1	2024-02-12	-0.0066
	1	2024-02-13	-0.0134
	1	2024-02-14	0.0107
	1	2024-02-15	0.0081
	1	2024-02-16	-0.0058
	1	2024-02-20	-0.0099

	portfolio_id	date	portfolio_daily_return
	2	2024-01-26	0.0040
	2	2024-01-29	0.0062
	2	2024-01-30	0.0104
	2	2024-01-31	-0.0146
	2	2024-02-01	0.0068
	2	2024-02-02	0.0031
	2	2024-02-05	-0.0037
	2	2024-02-06	0.0035
	2	2024-02-07	0.0071
	2	2024-02-08	0.0159
	2	2024-02-09	-0.0046
	2	2024-02-12	0.0049
	2	2024-02-13	-0.0105
	2	2024-02-14	0.0075
	2	2024-02-15	0.0104
	2	2024-02-16	-0.0058
	2	2024-02-20	-0.0003

	portfolio_id	date	portfolio_daily_return
	1	2024-02-21	0.0009
	1	2024-02-22	0.0316
	1	2024-02-23	-0.0044
	1	2024-02-26	-0.0030
	1	2024-02-27	-0.0014
	1	2024-02-28	-0.0056
	1	2024-02-29	0.0071
	1	2024-03-01	0.0054
	1	2024-03-04	-0.0142
	1	2024-03-05	-0.0150
	1	2024-03-06	-0.0009
	1	2024-03-07	0.0153
	1	2024-03-08	-0.0074
	1	2024-03-11	0.0004
	1	2024-03-12	0.0165
	1	2024-03-13	-0.0051
	1	2024-03-14	-0.0002

	portfolio_id	date	portfolio_daily_return
	2	2024-02-21	0.0012
	2	2024-02-22	0.0113
	2	2024-02-23	0.0007
	2	2024-02-26	-0.0039
	2	2024-02-27	0.0066
	2	2024-02-28	-0.0004
	2	2024-02-29	0.0044
	2	2024-03-01	0.0094
	2	2024-03-04	-0.0003
	2	2024-03-05	-0.0083
	2	2024-03-06	-0.0008
	2	2024-03-07	0.0066
	2	2024-03-08	-0.0057
	2	2024-03-11	0.0057
	2	2024-03-12	0.0021
	2	2024-03-13	0.0031
	2	2024-03-14	0.0008

	portfolio_id	date	portfolio_daily_return
	1	2024-03-15	-0.0059
	1	2024-03-18	0.0147
	1	2024-03-19	0.0055
	1	2024-03-20	0.0108
	1	2024-03-21	-0.0049
	1	2024-03-22	0.0038
	1	2024-03-25	-0.0025
	1	2024-03-26	-0.0017
	1	2024-03-27	0.0048
	1	2024-03-28	-0.0035
	1	2024-04-01	0.0028
	1	2024-04-02	-0.0130
	1	2024-04-03	0.0041
	1	2024-04-04	-0.0078
	1	2024-04-05	0.0096
	1	2024-04-08	0.0031
	1	2024-04-09	0.0033

	portfolio_id	date	portfolio_daily_return
	2	2024-03-15	-0.0028
	2	2024-03-18	0.0156
	2	2024-03-19	0.0042
	2	2024-03-20	0.0144
	2	2024-03-21	0.0084
	2	2024-03-22	-0.0012
	2	2024-03-25	0.0035
	2	2024-03-26	0.0009
	2	2024-03-27	0.0040
	2	2024-03-28	0.0047
	2	2024-04-01	0.0033
	2	2024-04-02	-0.0026
	2	2024-04-03	0.0023
	2	2024-04-04	-0.0152
	2	2024-04-05	0.0114
	2	2024-04-08	-0.0008
	2	2024-04-09	0.0024

Query #2 (The Best-Performing Sector Over Time)

```
CREATE OR REPLACE VIEW view_sector_performance AS
WITH latest_returns AS (
  SELECT
    symbol,
    cumulative_return,
    date
  FROM view_cumulative_returns
  WHERE (symbol, date) IN (
    SELECT symbol, MAX(date)
    FROM view_cumulative_returns
    GROUP BY symbol
  )
),
sector_returns AS (
  SELECT
    s.sector,
    lr.cumulative_return
  FROM latest_returns lr
  JOIN stocks s ON lr.symbol = s.symbol
)
SELECT
  sector,
  ROUND(AVG(cumulative_return), 4) AS avg_sector_return
FROM sector_returns
GROUP BY sector
ORDER BY avg_sector_return DESC;
```

	sector	avg_sector_return
▶	Technology	1.4997
	Consumer Discretionary	1.2892
	Communication Services	1.2671
	Financial	1.1667
	Consumer Staples	1.1262
	Energy	1.1064
	Automotive	1.0459
	Healthcare	0.9521

Query #3 (Stocks with the Lowest Risk)

```
CREATE OR REPLACE VIEW view_stock_risk AS
WITH stock_returns AS (
  SELECT
    symbol,
    date,
    (close_price - LAG(close_price) OVER (PARTITION BY
symbol ORDER BY date)) /
    LAG(close_price) OVER (PARTITION BY symbol ORDER BY
date) AS daily_return
  FROM historical_returns
)
SELECT
  symbol,
  ROUND(STDDEV_SAMP(daily_return), 6) AS volatility
FROM stock_returns
WHERE daily_return IS NOT NULL
GROUP BY symbol
ORDER BY volatility ASC;
```

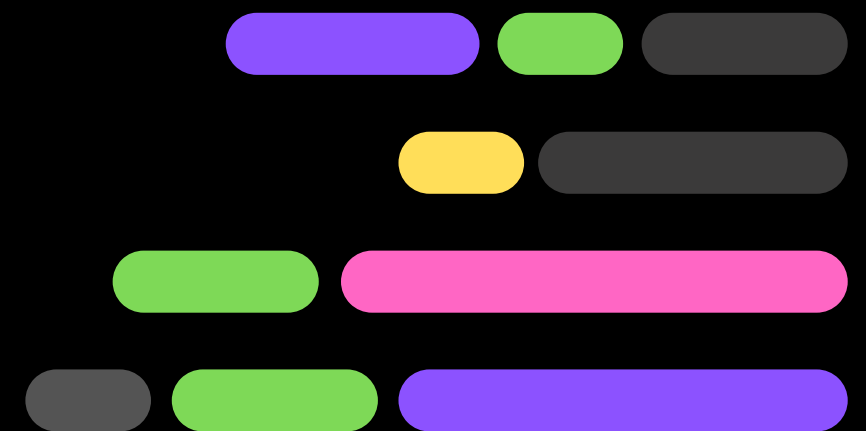
	symbol	volatility
▶	BRK-B	0.007456
	WMT	0.010564
	PEP	0.010691
	XOM	0.01137
	CVX	0.011414
	JPM	0.011853
	MSFT	0.011953
	BAC	0.012934
	UNH	0.015248
	AAPL	0.015258
	AMZN	0.01577
	GM	0.016188

	symbol	volatility
	AAPL	0.015258
	AMZN	0.01577
	GM	0.016188
	GOOG	0.017585
	GOOGL	0.017867
	DIS	0.018351
	NFLX	0.019791
	META	0.026456
	AVGO	0.026722
	NVDA	0.032962
	TSLA	0.033443

Analysis, Trends, & Insight

Based on queries:

- Based on the query results using cumulative returns grouped by sector (query #2), the best sector to invest in is technology
- Additionally, based on the query results using daily return to calculate risk (query #3), the stock with the lowest risk is BRK-B = Berkshire Hathaway
- Based on both queries, the technology stock with the lowest risk is MSFT = Microsoft

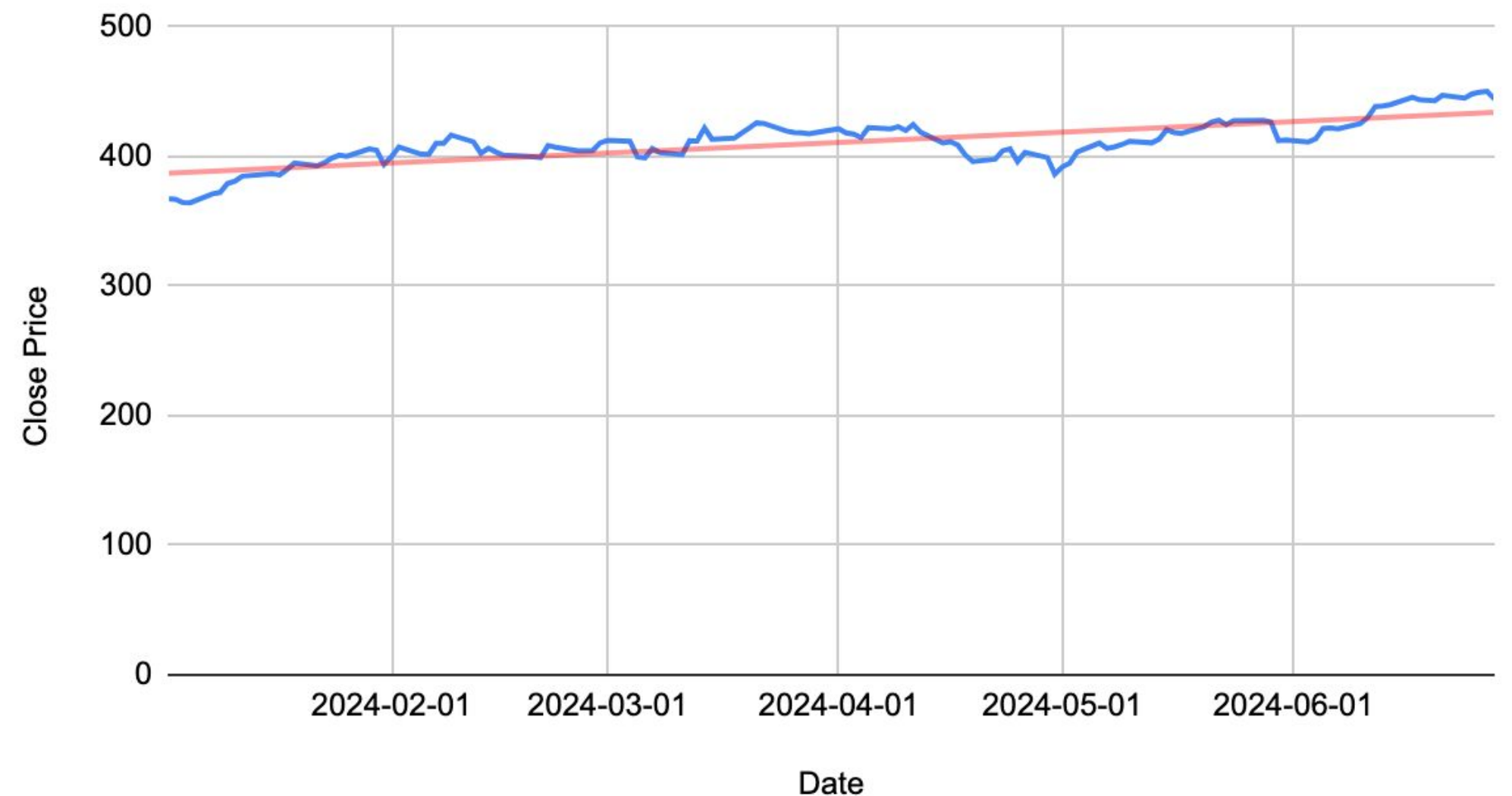




Trends - 1/6 - Closing Price

Microsoft has been steadily increasing, is in the best performing sector, and is also one of the stocks with the lowest risk.

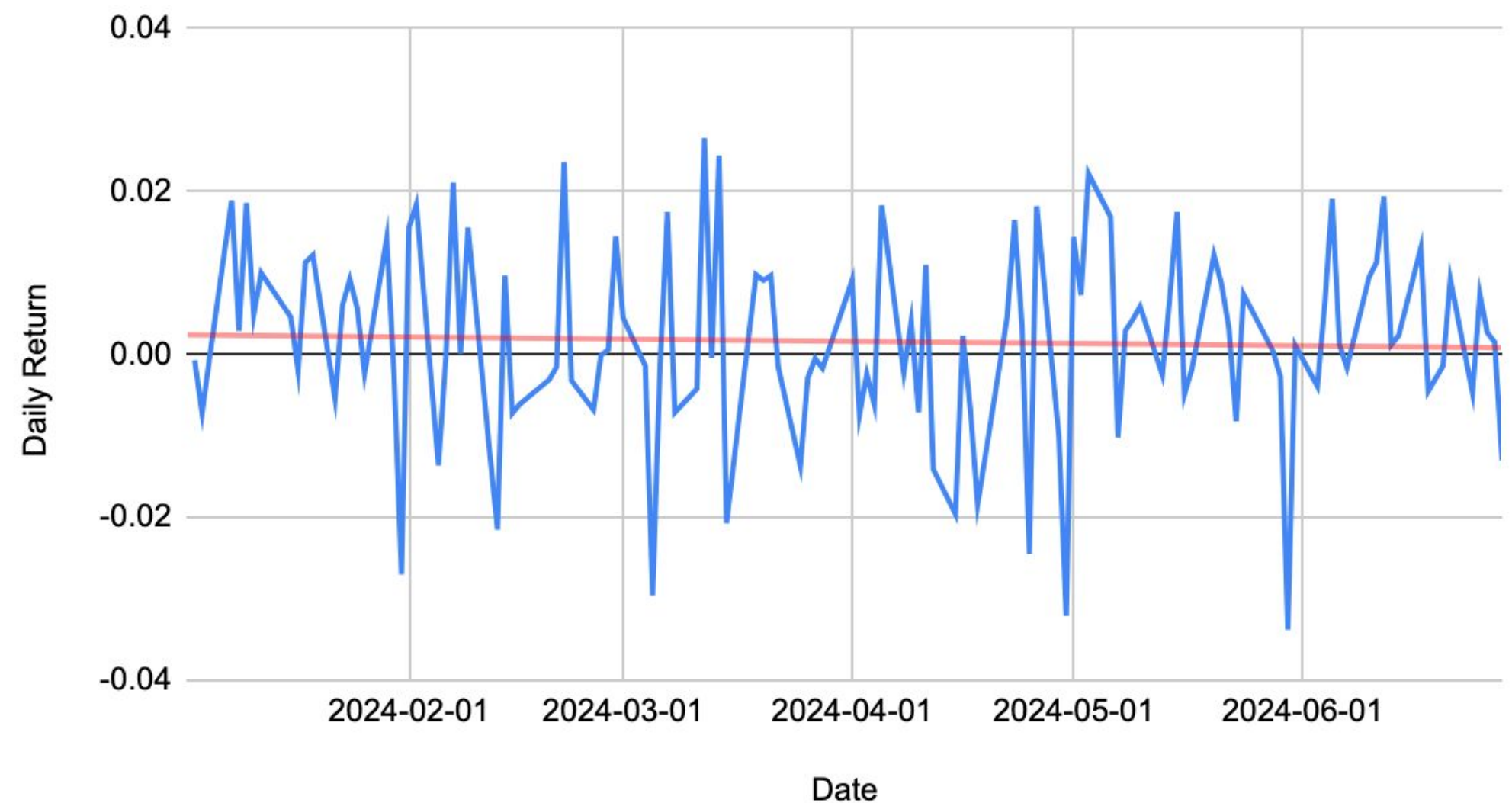
Microsoft Corp.



Trends - 2/6 - Daily Returns

Stocks tend to be more volatile when compared on a day to day basis.

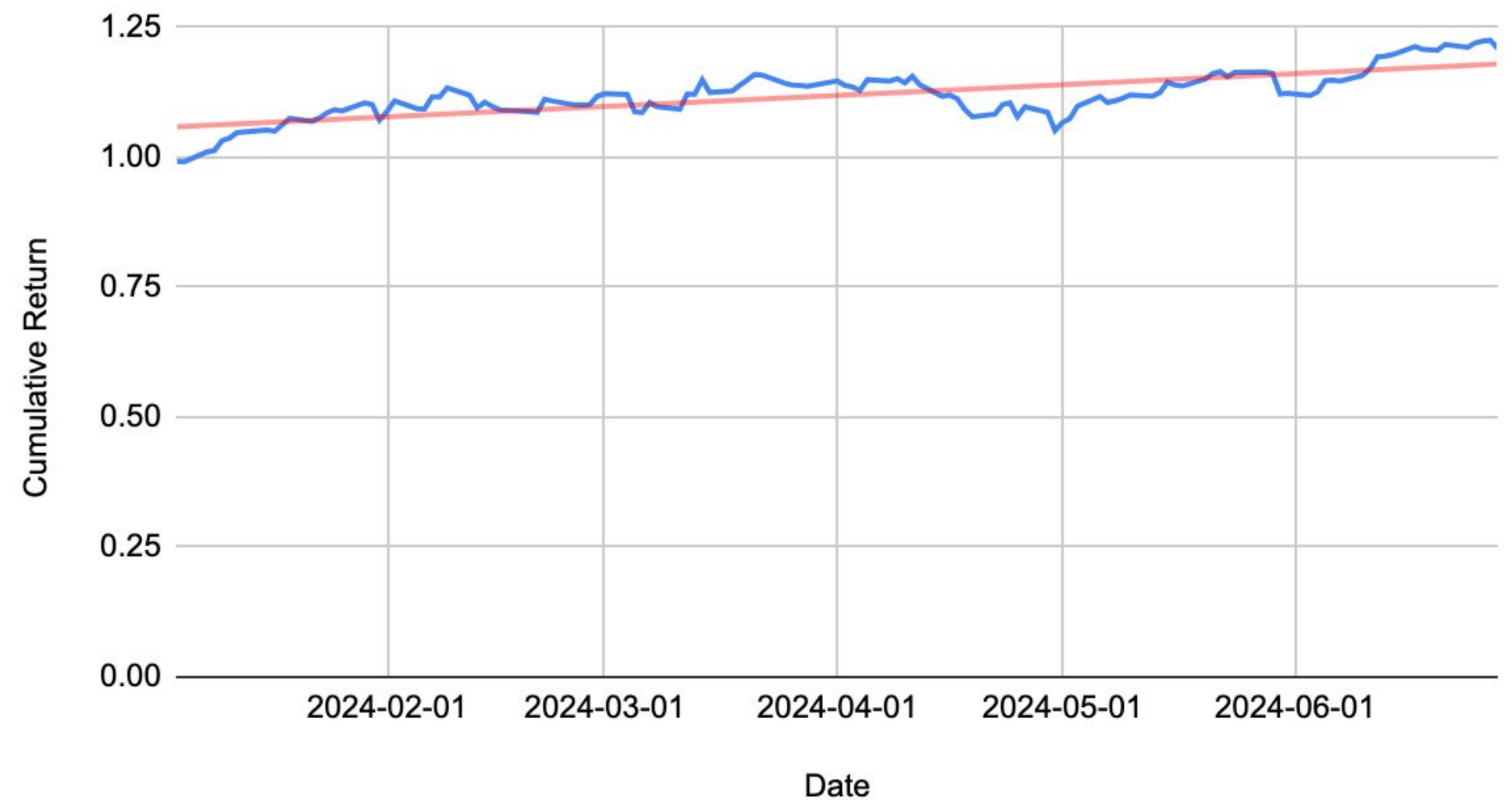
Microsoft Corp.



Trends - 3/6 - Cumulative Returns

The cumulative return for Microsoft has been increasing at a steady rate.

Microsoft Corp.

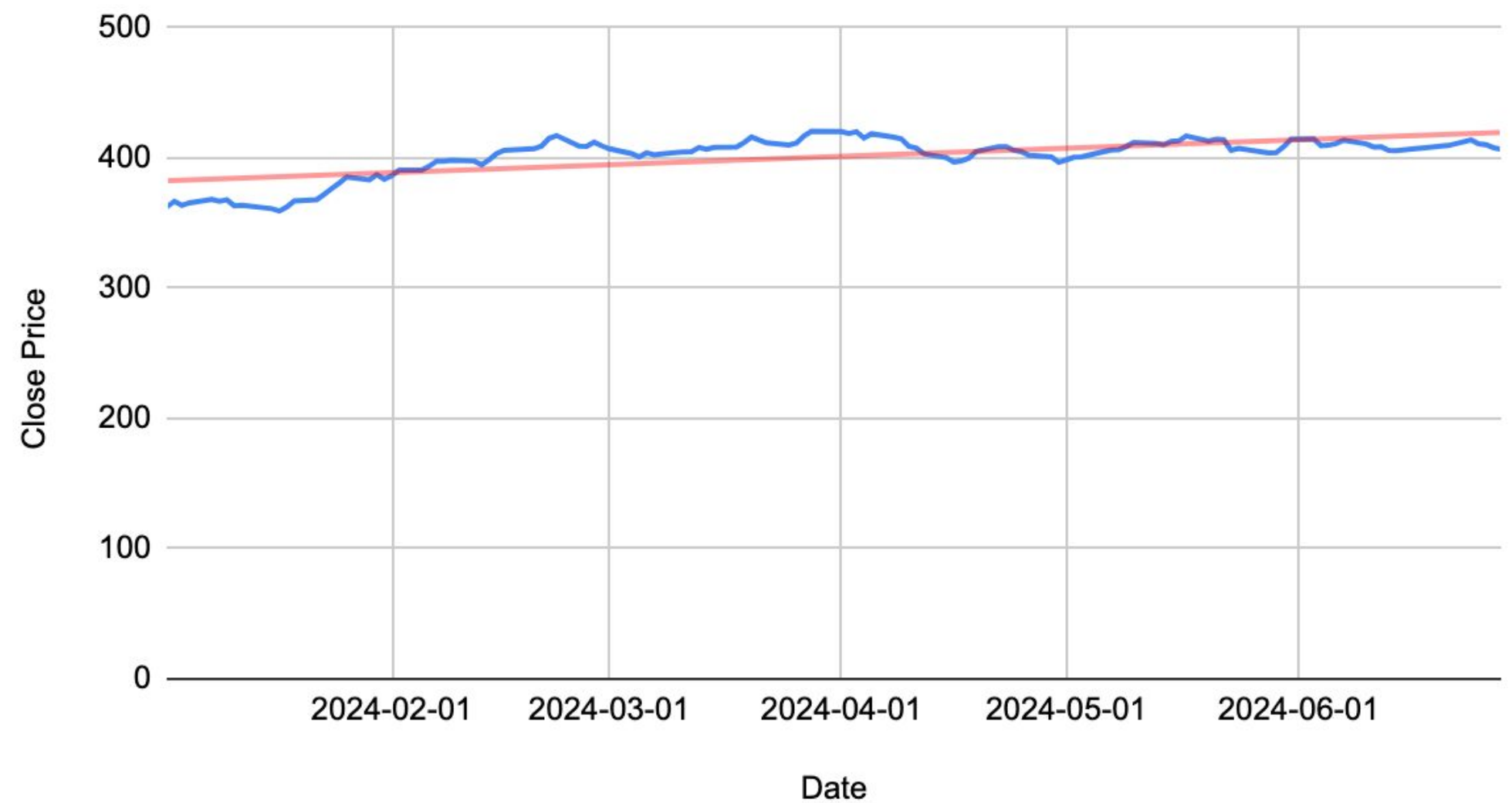


Trends - 4/6 - Closing Price

Berkshire Hathaway Inc. is the company with the lowest calculated risk.

Many people trust Warren Buffett's reputation but also, investing in Berkshire Hathaway is like investing in a portfolio in that it encompasses/holds many companies. It has been historically trusted as a safe investment

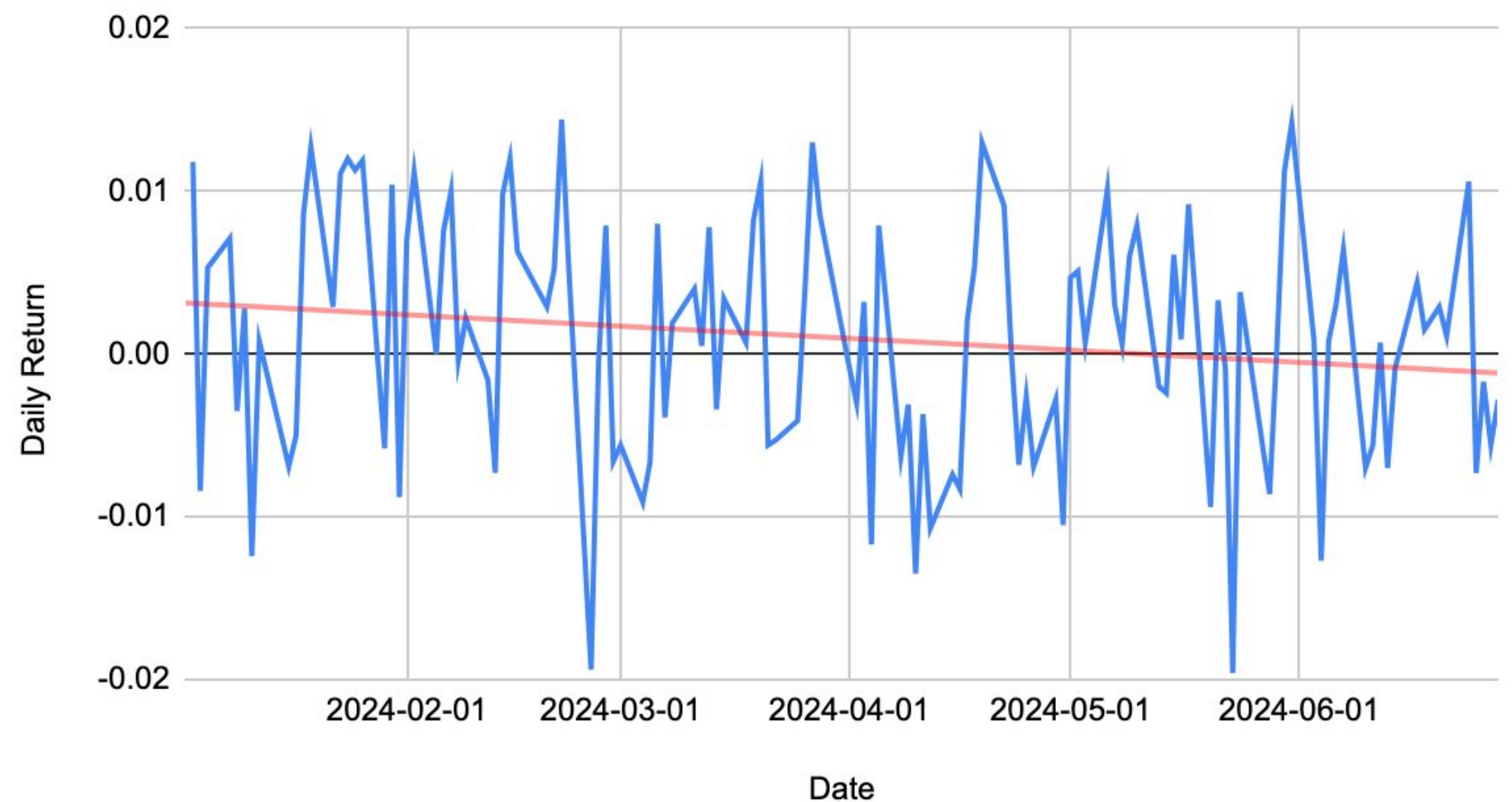
Berkshire Hathaway Inc.



Trends - 5/6 - Daily Returns

Like Microsoft, Berkshire Hathaway also has volatile daily returns. It would be a better choice for a long-term investment rather than a short-term investment.

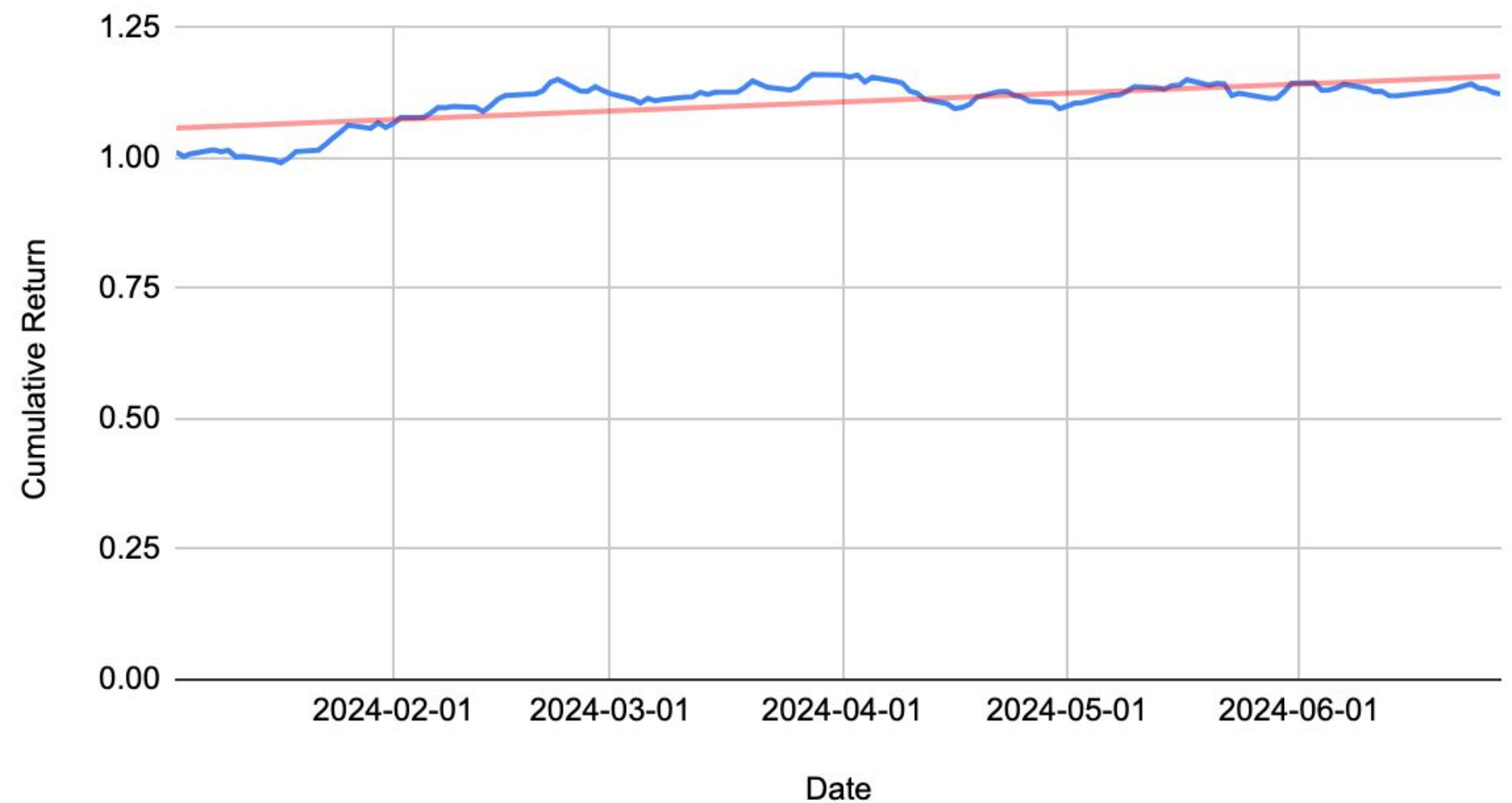
Berkshire Hathaway Inc.



Trends - 6/6 - Cumulative Returns

Berkshire Hathaway's stock has been steadily increasing in cumulative returns.

Berkshire Hathaway Inc.

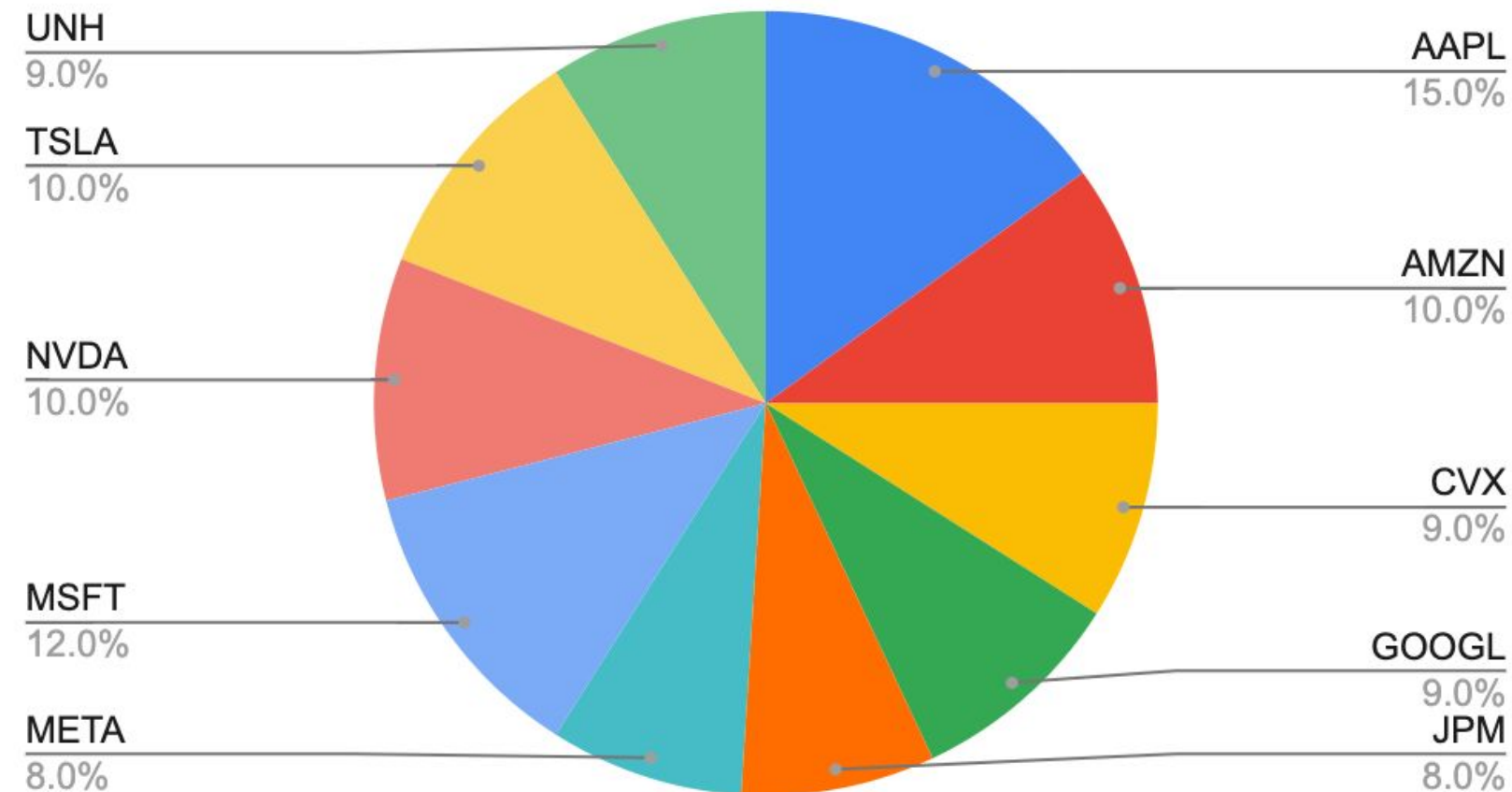




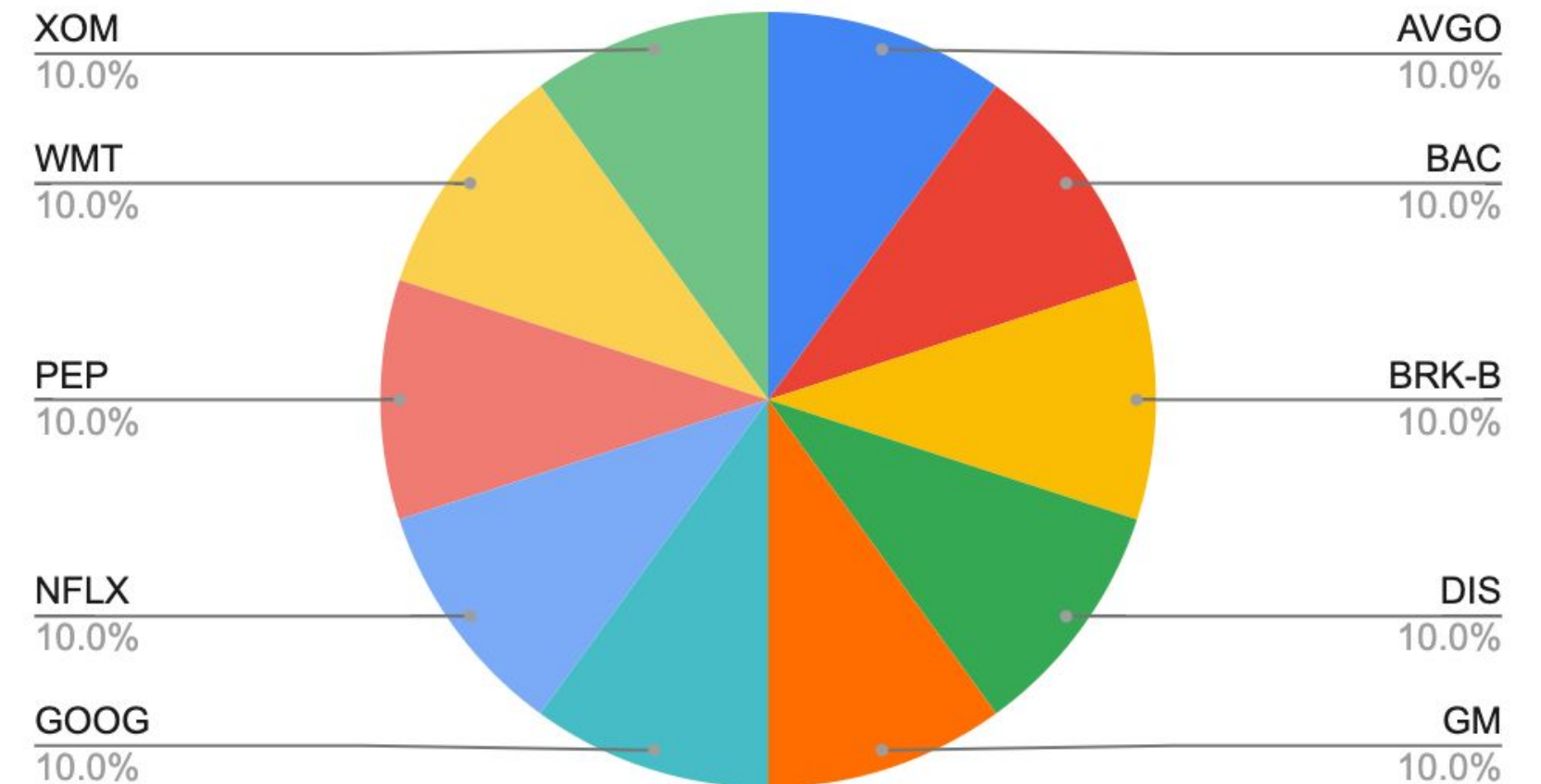
Portfolio Comparison - 1/2

Portfolio 1 is mostly investing in technology companies while Portfolio 2 has a larger, more even distribution.

Portfolio 1 Allocation Percentages



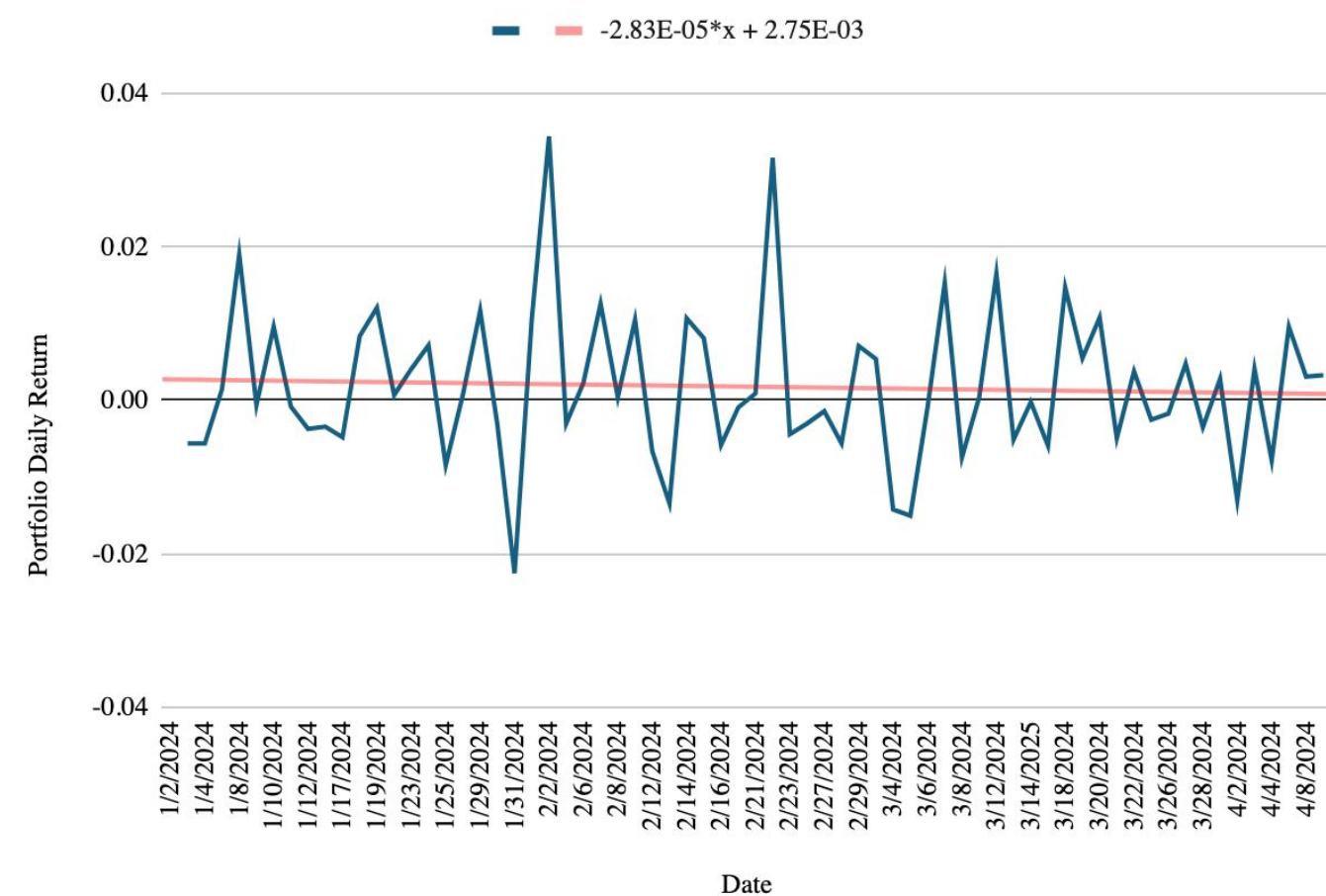
Portfolio 2 Allocation Percentages



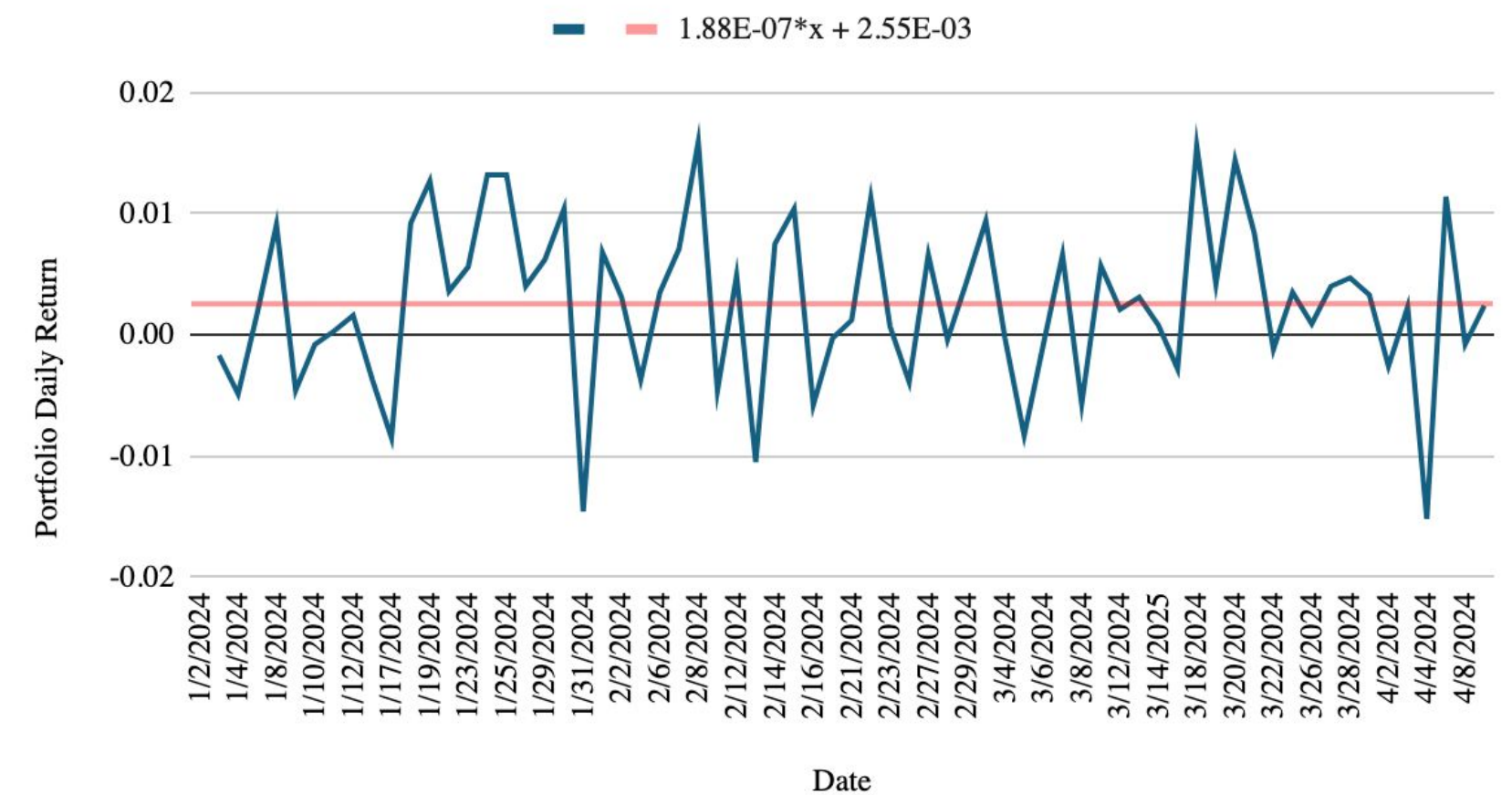
Portfolio Comparison - 2/2

The portfolio that was the most profitable was: **Portfolio 2**

Portfolio 1



Portfolio 2



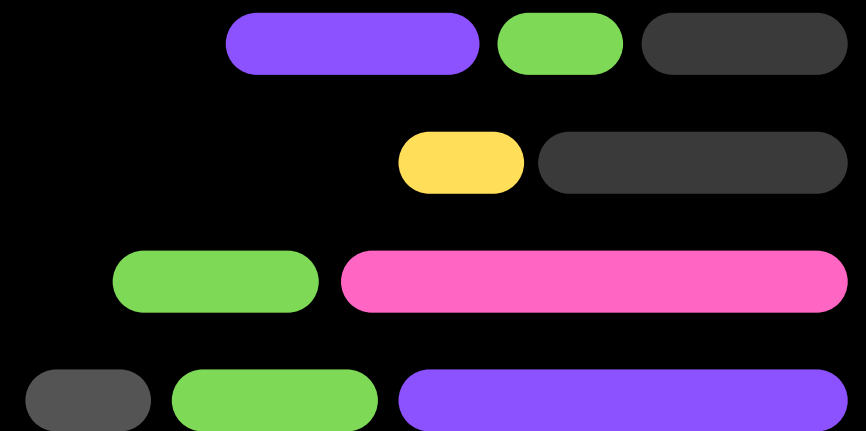
Extra Improvements

Table Changes:

- Removed stock_id and replaced it with symbol as the primary key in all tables
- Removed return_id and stock_id and created a composite primary key consisting of symbol and date for historical_returns table
- In the historical_returns table, the cumulative_returns column and daily_returns column were removed and can instead be accessed through a view

Other:

- Future database expansion could involve a investor table to hold investor information as well as which portfolio each investor is investing in, how much money, etc.



THANK YOU!

Any Questions?

